DEM-Based LOS Modeling: Mapping Satellite Dead-zones

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With special thanks to PhD student Misa Toman and Maricopa County SAR Jason Swencki

Introduction

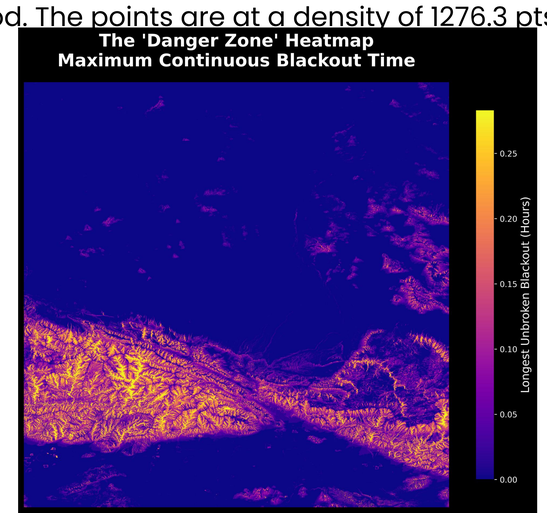
Integration of satellite connectivity into consumer hardware has improved backcountry safety but introduces additional considerations when used in extreme terrain. Current geodesic models within consumer satellite devices fail to account for local topographic occlusion from extreme terrain, creating a false sense of security. This project presents a high-performance framework, **heatmapgpu.py.** The Python code performs deterministic Line-of-Sight (LoS) calculations by fusing high-resolution USGS 3DEP DEMs and TLE orbital data to create high resolution maps showing relevant data of the quality and duration of satellite connections over a seven day time period.

Outputs

The code is modular and can easily use it's results to create different analysis for different use cases. Here are two examples showing significant use case using the Apple Globalstar network.

(Right) The analysis of the Devils Backbone trail within Angeles National Forest shows both the elevation profile as well as the coverage. At lower elevation, satellite coverage is occluded to a significant (~40%) percentage of the 168 hour period.

(Left) Each dot on the map is colored according to the maximum continuos blackout time over the 168 hour period. The points are at a density of 1276.3 pts/ km^2.



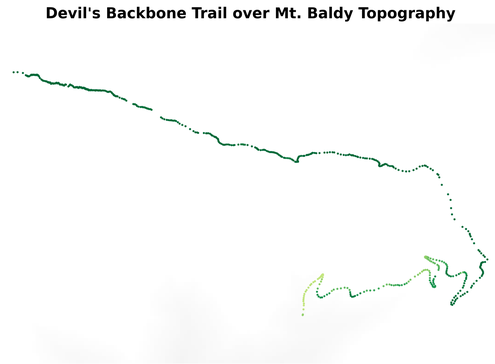
Methods

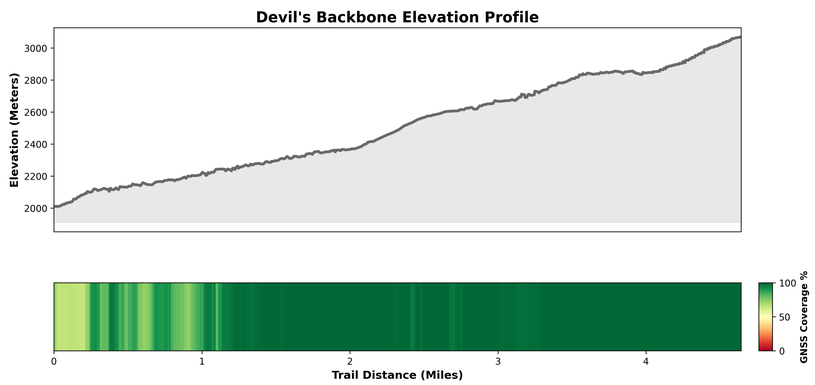
The development of the satellite occlusion Python simulation engine progressed through a technical hierarchy of increasing specialization to address the performance needs of the project. Initial development utilized a CPU-based framework leveraging Skyfield, Rasterio, and GeoPandas to validate core Line-of-Sight (LoS) logic.

To manage the significant computational overhead of high-fidelity 7-day simulations for large constellations, the methodology transitioned from local multithreaded processing via the joblib library to massively parallel C++ vector calculations optimized for NVIDIA A100 GPU servers.

The current architecture employs custom C++ CUDA kernels compiled via CuPy, which performs GPU-native ECEF-to-ENU coordinate transformations and compares satellite elevation against pre-calculated horizon masks.

This shift to kernel-level optimization resulted in a *1000x* increase in throughput over initial CPU-based implementations.





Discussion

Practical accuracy was verified through field testing in the Mt. Baldy – Angeles National Forest region, where the simulation successfully identified signal losses that geodesic-based consumer iPhone 14 Pro satellite communication application failed to detect.

Testing focused on the transition period the simulation predicted a status change from connected to blocked with hopes to qualitatively measure the refractive effect of the local granite peaks.

At the Manker Flats Trailhead, the iPhone UI generated a dangerous "False Positive" by predicting pending connection at 6:32 PM; however, the ray-tracing engine correctly flagged a continuous blackout caused by ridge obstruction.

Conversely, testing at the Mt. Baldy Ski Lift identified the model's operational boundaries, where man-made steel structures may have caused signal interference not accounted for by the terrain-only Digital Elevation Model (DEM).

These point-specific validations confirm that a high-fidelity terrain model provides more reliable predictions for Search and Rescue (SAR) environments than standard geometric models.

The following visualizations scale these point-tests into regional analytics, translating millions of individual GPU-accelerated calculations into actionable safety metrics for the San Gabriel Mountains.

Acknowledgments

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